USB PD CTS ENGINEERING CHANGE NOTICE

NOTICE: Any Company or Companies submitting a USB Power Delivery ECR proposal must be one of the following: a Promoter or Contributor of the USB 3.0 and 2.0 Specifications who have completed the USB Power Delivery addendum. If a group of Companies is submitting an ECR proposal, each company must be either a Promoter or Contributor of the USB 3.0 and 2.0 Specifications who have completed the USB Power Delivery addendum.

SPECIFICATION REVISIONS AND ADDENDA: At any point in time, there shall only be one current version of the USB PD CTS, termed the production version. At the same time, there may also be proposed revisions to the specification's design which are not yet approved and shall be held confidential as deemed necessary by the USB 3.0 and USB 2.0 Promoters and within the Group of Working Committee(s).

PROCEDURES FOR SUBMITTING PROPOSALS: Both members of the USB Implementers Forum as a whole and members of the USB 3.0 and USB 2.0 Promoters may submit requests to revise the USB PD CTS Specification. Such a request may be rejected or may result in a USB PD Engineering Change Notice (ECN), which is the official way USB specifications may be changed.

FORMAT OF PROPOSAL: The originator of a request to alter the USB PD CTS Specification may do so by posting this to the USB Power Delivery Compliance working group for review. Once the proposal has been reviewed by the working group it will be passed to the USB 3.0 and 2.0 Promoters for approval to publish.

RESUBMISSION AND APPEAL: The originator of a request that was not approved can redraft the original request. Rewritten proposal will be treated as a new proposal and will be evaluated using the procedures described above. The originator of a request that was not approved can also submit an appeal to the USB 3.0 and 2.0 Promoters. The appeal must be made in writing and addressed to the Secretary of the USB Implementers Forum.

ABOUT THE ENGINEERING CHANGE REQUEST FORM:
The Purpose of this Engineering Change Request Form is to expedite the review process of the proposal by providing explanations, background information, and examples of the proposed changes at a high level. This form serves as an executive summary to the actual proposal.

STEPS ON HOW TO SUBMIT A USB PD ENGINEERING CHANGE REQUEST:

1) Please fill out the Engineering Change Request Form on the following pages completely:
   a) Detail the names and contact details for each of the ECR contributors
   b) Update the ECR Title
   c) Give a minimum of 2-3 sentences for each description on the form outlining the background to the ECR
2) For each section/table/figure to be updated:
   a) Detail the section number, starting page and figure/table number to be updated as appropriate.
   b) Detail existing text under "From Text"
   c) Detail changed text under "To Text"
3) Save the file as "USB PD CTS 1.0 R 1" followed by the ECR Title as per step 1)b)
4) Post the ECR in the USB PD CTS Documents section under "ECR | New ECRs".
   a) This ECR will then be reviewed by the Power Delivery Compliance Working Group.
   b) Revisions to the ECR originating from the review should be submitted as document revision of the original ECR using "Add new document".

USB PD CTS ENGINEERING CHANGE NOTICE

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USB PD CTS ENGINEERING CHANGE NOTICE

Title: TEST.PD.PROT.SNK3.5 Sink PPS Normal Operation
Applied to: USB PD CTS Specification Q3 2026

Brief description of the functional changes proposed:

Update the RDO object-position check in [TEST.PD.PROT.SNK3.5#2] so the Object Position field is expected to reference the PPS APDO in the last Source_Capabilities Message sent by the Tester, rather than a static object position value.

Background Information:
COMMON.PROC.BU.5 was recently updated to better align with USB PD Source power rules by adding three PDOs, increasing the advertised PDO count from two to five:

Original	5V Fixed	-	-	-	PPS APDO 5V-21V
Updated	5V Fixed	9V Fixed	15V Fixed	20V Fixed	PPS APDO 5V-21V

As a result, the static RDO object-position check in TEST.PD.PROT.SNK3.5#2 incorrectly expects the Object Position field to be 2 instead of 5.

Benefits as a result of the proposed changes:

- PPS-capable Sinks can successfully pass TEST.PD.PROT.SNK3.5.
- The dynamic wording remains valid if the bring-up procedures change their advertised PDOs again.

An assessment of the impact to the existing revision and systems that currently conform to the USB specification:

None

An analysis of the hardware implications:

None

An analysis of the software implications:

None

An analysis of the compliance testing implications:

Tester software must be updated.

An analysis of the Vendor Info File (VIF) implications:

None

USB PD CTS ENGINEERING CHANGE NOTICE

Actual Change Requested

(a). TEST.PD.PROT.SNK3.5 Sink PPS Normal Operation, Page 181

From Text:

Test Procedures:

1. There are 2 possible bring-up procedures:
    a. The UUT has VIF field PD_Port_Type set to Provider/Consumer, the Tester runs bring-up procedure with the UUT as a Sink COMMON.PROC.BU.12
    b. The UUT has VIF field PD_Port_Type set to anything else, the Tester runs bringup procedure with the UUT as a Sink COMMON.PROC.BU.5
2. The Tester presents *SinkTxOk*.
3. The check fails if a *Request* Message is not received within *tPPSRequest* max. [TEST.PD.PROT.SNK3.5#1] The delay is measured from the time the last bit of the EOP of the *GoodCRC* Message sent by the Tester in response to the previous *Request* Message to the time the first bit of the Preamble of the new *Request* Message is received.
4. The Tester runs the following additional field checks on the Request Data Object: [TEST.PD.PROT.SNK3.5#2]
    a. Object Position field is 2.

To Text:

Test Procedures:

1. There are 2 possible bring-up procedures:
    a. The UUT has VIF field PD_Port_Type set to Provider/Consumer, the Tester runs bring-up procedure with the UUT as a Sink COMMON.PROC.BU.12
    b. The UUT has VIF field PD_Port_Type set to anything else, the Tester runs bringup procedure with the UUT as a Sink COMMON.PROC.BU.5
2. The Tester presents *SinkTxOk*.
3. The check fails if a *Request* Message is not received within *tPPSRequest* max. [TEST.PD.PROT.SNK3.5#1] The delay is measured from the time the last bit of the EOP of the *GoodCRC* Message sent by the Tester in response to the previous *Request* Message to the time the first bit of the Preamble of the new *Request* Message is received.
4. The Tester runs the following additional field checks on the Request Data Object: [TEST.PD.PROT.SNK3.5#2]
    a. Object Position field references the PPS APDO in the last Source_Capabilities Message sent by the Tester.